ADVANCED INTELLIGENT SYSTEMS
www.advintellsyst.com

BeeRootBot: A Bioinspired Robotic Probe Exhibiting Apical Growth through In Situ Soil Binding

Sachin Sachin, Alessio Mondini, Stefano Mariani, Emanuela Del Dottore, and Barbara Mazzolai**

This study introduces a minimally invasive robotic probe inspired by plant root growth, designed for subsoil exploration and future ecosystem monitoring and intervention. The bio-inspired probe advances in soil by mimicking plant root apical growth, creating and consolidating a borehole through the injection of a bio-based, biodegradable binder at its tip. This innovative process confines penetration resistance to the tip while generating a hollow tubular structure by harnessing in situ local soil. The probe's penetration is facilitated by a linear actuator, which can be retracted upon reaching a desired depth, thereby minimizing the environmental dispersion of mechatronic components. This approach not only enhances the efficiency of subsoil exploration (whether on-Earth or in outer space) by reducing penetration force requirements and reliance on exogenous material but also ensures environmental sustainability by employing biodegradable materials and lowering mechanical footprints. The robotic probe's design and functionality highlight the potential of bio-inspired technologies to address complex environmental challenges, paving the way for future innovations in ecological research and conservation efforts. This study underscores the importance of integrating biological principles into engineering solutions to develop tools that are both effective and environmentally responsible.

1. Introduction

The exploration of subterranean environments presents significant challenges due to unstructured soil textures, unpredictable impenetrable obstacles, and variable high soil impedance.[1] Drilling and boring are the primary practices in this domain.[2] However, these processes involve significant initial setup costs and extensive soil excavation work and sometimes involve a two-step process to excavate and stabilize the borehole. For on-Earth soil exploration, less invasive solutions should be prioritized to minimize disturbances to fragile ecosystems, particularly in applications related to soil environmental monitoring and intervention. Similarly, in outer space missions, reducing invasiveness is crucial for preserving the integrity of the environment being explored. In both scenarios, minimizing disruption, material usage, and time can significantly reduce the overall mission costs.

With advances in robotics and automation, researchers are investigating new small-scale, less invasive, and possibly cost-effective technologies,[3,4] such as burrowing mechanisms,[5–11] peristaltic motion-based systems,[12–18] moling techniques,[19,20] and growing systems.[21,22] Among these technologies, growing mechanisms represent a recently explored new class of robots that imitate plant-like apical growth by adding seed-feedstock material at their extremity.[23] This movement strategy has been shown to facilitate soil penetration, as demonstrated through both experimental[24] and numerical[25] investigations. Due to this feature, plant-inspired growing mechanisms have been actively investigated for subterranean exploration[22,26–29] and for enabling access to soil monitoring systems.[30] In the subterranean region, the pathway for such advancing mechanisms is fully packed with soil media. To overcome the resistive forces encountered during advancement, the moving system requires a powerful actuation mechanism, which, in turn, results in relatively high-energy consumption.[31] A tethered power source becomes a viable solution to meet such high-energy demands. However, the tethered energy line further requires an accessible passage to move freely within the subterranean environment. Retaining this passage presents another challenge, as the soil media is prone to collapse.

In this respect, growing systems that also construct a trailing support body, such as those based on additive manufacturing,[27,32] utilize the built structure to provide a pathway for the power line while simultaneously enabling spatial advancement. However, these systems leave their structural body behind in the environment. As they grow longer, more feedstock material is required and is subsequently disposed of afterward. In outer space missions, minimizing material usage is crucial to save space and reduce costs. For on-Earth applications, the use of biocompatible and biodegradable materials for growth is an important consideration to enhance sustainability.

S. Sachin, A. Mondini, S. Mariani, E. Del Dottore, B. Mazzolai
Bioinspired Soft Robotics Lab
Istituto Italiano di Tecnologia
16163 Genova, Italy
E-mail: sachin.sachin@iit.it; barbara.mazzolai@iit.it

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/aisy.202500720.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

DOI: 10.1002/aisy.202500720

This study investigates the possibility of harnessing in situ soil as a direct feedstock material for the automated borehole lining process performed by an apically growing robotic system. The primary objective is to reduce the amount of exogenous material brought into the environment by employing local resources, i.e., the soil. However, binder material is still required to reinforce the subsoil passage. Consequently, our secondary objective is to create a continuous subsoil channel featuring hydrophobic, biocompatible, and biodegradable characteristics. A hydrophobic channel can effectively sustain changes in moisture levels without compromising its strength, while biocompatibility and biodegradability are prioritized to respect and safeguard the subsoil ecosystem.

3D printing is actively being explored for mold-making in sand casting.[33–35] However, to facilitate sand reclamation, the binders used in the process are water soluble and therefore unsuitable for creating water-resistant structures.[36] Additionally, commercial binders such as furans and phenolic resins are toxic and thus not ideal for the subsoil ecosystem.[37] Inorganic eco-friendly binders, such as sodium silicates, are hygroscopic and do not meet the initial design requirement for a water-resistant structure.[38] Rapid curing is another desired property of the binder material for growing and facilitating extrusion. To this end, we selected beeswax as a bio-based binder material. Beeswax is derived from living organisms and is composed of various organic compounds, primarily esters of fatty acids and long-chain alcohols. Beeswax is biodegradable; according to scientific literature, it has been found to be readily biodegradable in standard tests, i.e., achieving >60% biodegradation within 28 days.[39] This allows the structure of the growing robots to be left in the ground once it has grown, without the need to retrieve it. Beeswax, being a wax, is insoluble in water, which allows the grown structure to withstand potential rain. In addition, the melting temperature of beeswax ranges from 60–65 °C[40]; reducing the impact on the subsoil environment and lowering the energy demand for melting the binder during extrusion.

The specific contribution of this study is the development of a minimally invasive, plant root-inspired growing probe named BeeRootBot for subsoil exploration and targeting future ecosystem monitoring and intervention applications. **Figure 1**a depicts a potential deployment scenario. The probe mimics apical growth to advance in soil by generating and consolidating a borehole by injecting a bio-based and biodegradable binder at the tip (Figure 1b). This process confines penetration resistance at the tip by forming a hollow tubular structure using the in situ local soil. Penetration is facilitated through a retractable linear actuator to minimize the dispersion of robotic components in the environment. The remainder of the article is organized as follows: the design and development of the growing system and probe are presented in Section 2.1 and 2.2, followed by structural characterization methods in Section 2.3. Then, a discussion on the penetration performance and monitoring application is provided in Section 2.4 and 2.5. Finally, Section 3 offers conclusive remarks and suggests future research directions.

2. Results

2.1. Bioinspiration and Design of the Growing System

Plant roots feature primary growth at the apical meristem region, where cell division and elongation occur (Figure 1b).[41] This biological phenomenon recalls an additive manufacturing process through which new cell material is added layer-by-layer at the

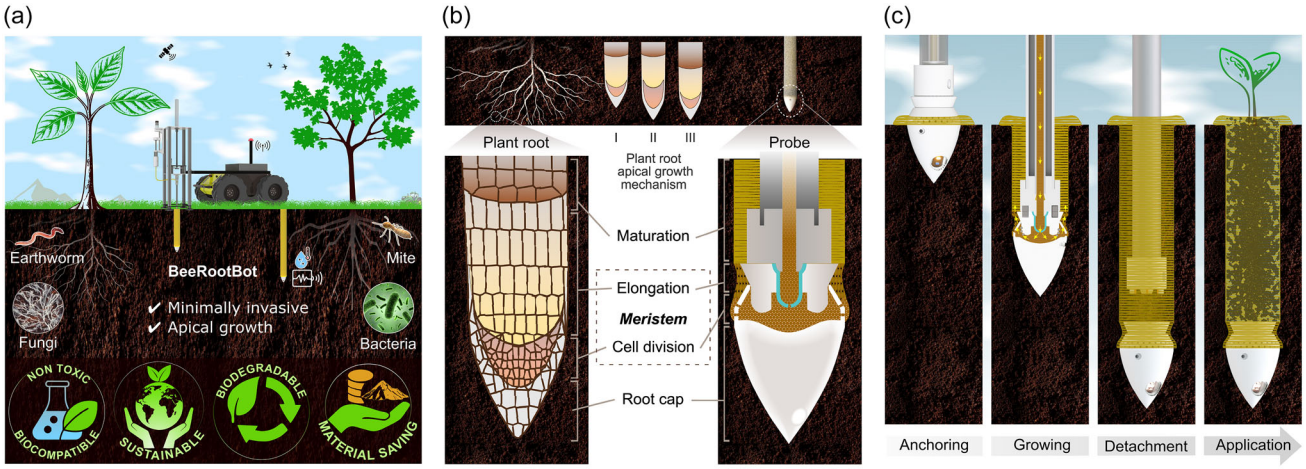

Figure 1. a) Envisioned scenario for BeeRootBot, a robotic growing probe that constructs its body by harnessing soil and binding it with natural beeswax to increase sustainability and reduce material usage. b) Schematic presenting the analogy between the apical region of a plant root and the BeeRootBot probe (section view). In plant roots, the primary growth occurs at the apical meristem through cell division (step II in the top-middle part) and elongation (step III in the top-middle part), resulting in the tip advancement (tip displacement visible from steps I and II in the top-middle part). The apical region is usually divided into four parts (bottom part: juxtaposition between natural and artificial apical regions). Analogous to plant roots, the probe has a rigid cap to house sensing elements. "Cell division" involves beeswax infusion into the soil, "elongation" involves axial displacement of soil-beeswax mix along the circumferential collar, and "maturation" involves beeswax solidification. c) Sketch depicting different process stages of artificial growth in subsoil: i) "anchoring", a brim of beeswax is built on the surface to prevent soil particles falling into the channel; ii) "growing", beeswax is extruded to form a channel inside soil (yellow arrows indicate the flow of beeswax). The tip pushes through the soil, forming a cylindrical hollow structure as the beeswax solidifies; iii) "detachment", the intruding shaft along with connector and supply channel are retracted; and iv) "application", monitoring via probe and usage of the constructed channel for intervention. A visual media presenting the different process steps is provided in Video S1, Supporting Information.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

apical extremity. BeeRootBot imitates this process of adding cell material by injecting liquid beeswax into the in situ local soil from its tip while advancing spatially. The beeswax diffuses among soil particles and while solidifying, binds them to form a cylindrical hollow structure. The probe advances in soil due to the pushing exercised by an intruding shaft connected to it, which is powered by an external actuation mechanism (**Figure 2**a). The intruder moves inside the hollow channel without entering into contact with its walls, thus avoiding bodily friction on the shaft. The overall robotic system is composed of the apical cap, also referred to as the probe in the following, through which the binder material is injected and which can host sensors; the intruding shaft to which the probe is connected; the constructed tubular structure; and an external frame that houses the power supplier, actuators, and material reservoir (Figure 2a). The reservoir is composed of a heating syringe pump in which beeswax (Happy Bee Brand, Amazon) is added in pellet form, melted, and generates a controlled flow through the connecting silicone tube (ID 4 mm, OD 6 mm, Saint-Gobain Versitec), which passes through the intruding shaft and reaches the supply-side connector (Figure 2b1). The cap is pushed forward by the intruding shaft in synchronization with the flow of the binder. To ensure flowability, a commercially available high-pressure stainless steel syringe pump (New Era pump systems) has been customized by replacing its plunger with a custom-made Rigid-10K plunger, featuring screw-based manual air release vents. The vents provide an escape passage for the trapped air while filling beeswax into the syringe pump from the plunger side of the barrel.

The metallic syringe pump is wrapped with resistance heating Nichrome (NiCr) wire (0.5 mm diameter). Also, the silicone supply tube contains a PTFE-wrapped NiCr wire inside it to maintain the temperature of molten beeswax. The NiCr wire runs forward and then loops back within the same tube; on the forward path, it is insulated with a PTFE sleeve to prevent short-circuiting with the return segment. The surface of the syringe pump barrel and the heating coils are wrapped with Kapton tape for electrical insulation to avoid direct contact with the Nichrome wire. To minimize the heat losses, the wire wrapped over the barrel is further isolated by a cover made of Rigid-10K material. The length (l_r) of Nichrome wire wrapped over the syringe pump was estimated using thermal energy balance, as follows

$$l_r = \frac{f_{hl}(m_s\, c\, \Delta T)}{I_{max}^2\, r\, t_{in}} \tag{1}$$

where, f_{hl} is the heat loss factor (taken as 1.3, to account ambient heat loss), m_s is the mass of the syringe pump, c is the specific heat capacity of stainless steel, ΔT is the temperature difference, I_{max} is the upper limit of current (2 A), r is the resistance per unit length (5.33 $\Omega\,\mathrm{m}^{-1}$), and t_{in} is the time needed to reach the initial thermally stable state. The wire wrapping near the syringe pump outlet was kept denser than the plunger end. For the estimated 3 m-long wire, 25 turns were wrapped around the syringe pump. The temperature of the syringe pump was set to 70 °C, and the outer temperature for the silicone tube was set to 60 °C. Two separate MOSFETs controlled in proportional-integral-derivative regulated the temperatures via a microcontroller (CYBLE-014008-00, Infineon Technologies). Three temperature sensors were mounted to monitor the beeswax temperature: one on the syringe, one along the silicone tube, and one at the tip close to the extrusion. Pumping and penetration are driven by distinct brushed DC motors integrated with external gear reduction mechanisms. The penetration

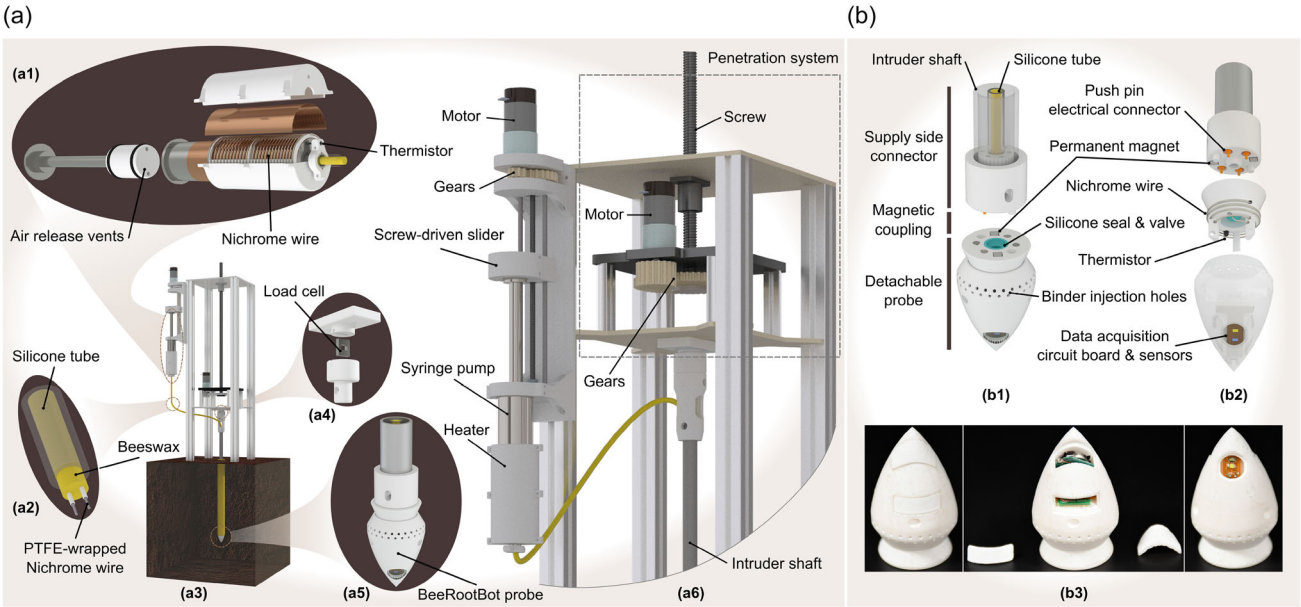

Figure 2. a) Detailed view (CAD rendered, NX 12.0, Siemens) of subsoil growing system and its components; a1) exploded view of syringe pump heating system assembly; a2) illustrated view of beeswax supply silicone tube, it carries a PTFE-wrapped Nichrome wire for maintaining the inside temperature; a3) growing system; a4) exploded view of uniaxial load cell assembly; a5) zoomed view of probe and its connector; and a6) magnified view of penetration and binder pumping system. b) Detailed view of BeeRootBot probe and its components; b1) magnetic coupling; b2) exploded view of the probe assembly; and b3) snapshots of the prototype probe showcasing the IC board and sensors used for remote subsoil monitoring.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

system also has a uniaxial load cell attached at the top of the intruding shaft. Additional details on the control of the system are provided in Text S1, Supporting Information.

2.2. Detachable Probe Design

The probe is made detachable to enable partial retraction and feature long-time remote monitoring capabilities in subsoil environment (Figure 2b). The probe is connected to the intruding shaft via a magnetic coupling, featuring both mechanical and electrical connections to ease detachment. The detachment can be performed by pulling the intruding shaft upward once the probe reaches the desired depth and the growing structure is cured enough to sustain disengagement resistance (Figure 1c). We experimentally observed that complete solidification is reached in ≈5 min, but even partial solidification is adequate to maintain structural integrity. To withstand high temperatures, the probe and its connectors are SLA 3D-printed using Rigid-10K resin. The connector of the probe houses a temperature sensor, a custom silicone seal and valve (Smooth-Sil 960), and a heating coil to keep the beeswax flowable. The silicone valve is designed to act as a pressure-operated outlet for the silicone tube, restraining the free flow of beeswax. Beeswax is infused into the soil through multiple injection holes positioned circumferentially along the major periphery of the probe (Figure 2b1). The probe has been designed to mimic the profile of *Zea mays* roots apex to reduce soil penetration resistance.[42] During penetration, for a specific soil, the contact pressure remains constant relative to the size of the penetrometer knob.[25] Therefore, a miniaturized probe is desirable to limit the magnitude of the penetration force. The prototyped probe reaches 3 cm in diameter at the prominent location (**Figure 3**a). The rest of the geometry is optimized iteratively by performing experiments with the objective of generating a continuous, consolidated stable channel inside soil (see Text S2, Supporting Information, for additional details).

The channel thickness is primarily governed by the difference between the radii of the probe root cap (r_r) and the connector (r_c) (Figure 3a). These parameters represent the dimensions of the extrusion die for the growing structure. The net flow of beeswax is channeled through this region. The overall thickness is a combination of the beeswax layer and soil particles and can be estimated as $\approx r_r - r_c$. We empirically search for a thickness that compromises beeswax consumption and the minimum amount of binder material required for a structurally reliable channel. We thus set an upper limit t_U based on the consumption of beeswax, restricting the structural composition to 80% beeswax by mass. The lower limit t_L was determined as the minimum amount of beeswax required to generate a viable growing channel. The specific grain size of our experimental soil (≤1 mm) was also considered in the analysis. A die with a cavity dimension that is too small will not provide enough space for the neighboring soil grains to get fully coated in beeswax, thereby resulting in poor binding. In our case, a die of dimension 2 mm provides an optimal balance between strength and binder savings (Figure 3a). The extent of beeswax infusion also affects the thickness; however, due to the high viscosity of beeswax, this parameter has a minor influence. The probe advancement rate (v_p) and beeswax flow rate of the syringe (v_s) are made synchronous and governed by the equivalence of volumetric flow rates of the syringe $\dot{V}_s$ and probe extrusion die $\dot{V}_p$ outlets. Being $\dot{V}_s$ described by

$$\dot{V}_s = v_s A_s \tag{2}$$

where, A_s is the cross-section area of the syringe pump barrel, and $\dot{V}_p$ expressed as

$$\dot{V}_p = v_p A_p = v_p \pi (r_r^2 - r_c^2) \tag{3}$$

where, A_p is the cross-section area of the extrusion die. The resulting syringe pump feed rate can be written as

$$v_s = \frac{v_p \pi (r_r^2 - r_c^2)}{A_s} \tag{4}$$

The penetration rate was then fixed upon the following empirical considerations (Figure 3b). A too-slow advancement rate (below the lower limit v_{p_L}) leads to slow heat dissipation into the soil due to prolonged heat transfer. This delays beeswax solidification, ultimately resulting in poor lining formation and excessive thermal energy consumption. On the other hand, an advancement rate exceeding the upper limit v_{p_U} does not provide sufficient time for the beeswax to solidify, compromising the quality of the lining and hindering proper infusion of the beeswax into the soil. For the current settings, an advancement rate of 0.5 mm s^{-1} provides a

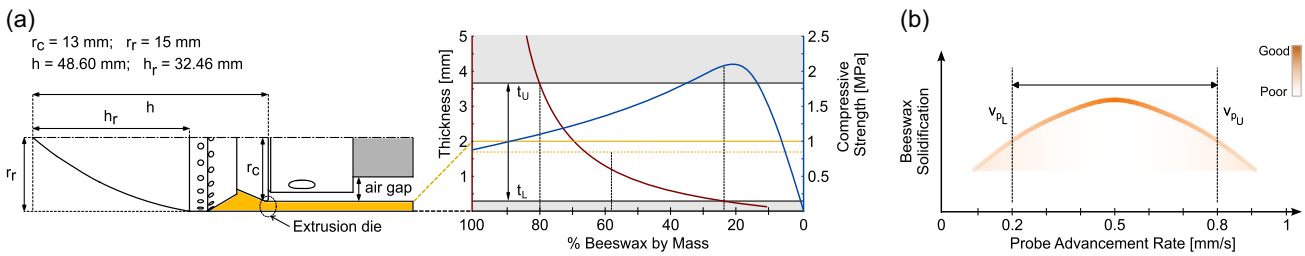

Figure 3. a) Probe design parameters and the relationship between extrusion die dimensions, beeswax consumption, and the strength of lining. The brown curve represents the expected composition of the growing channel with respect to the dimensions of the extrusion die. The blue curve represents the compressive strength of sand-beeswax composite for varying compositions, tests conducted as per ASTM D695. The upper and lower bounds (t_U and t_L) for thickness were determined by assessing the consumption of beeswax needed for viable channel growth. The continuous orange line represents the dimension of extrusion die, while the dashed orange line represents the observed thickness. b) Empirical evaluation of the relationship between probe advancement rate and beeswax solidification for a reliable channel construction.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

good compromise between energy consumption, solidification, and beeswax infusion.

2.3. Characterization and Structural Analysis

2.3.1. Sand-Beeswax Composite Strength Test

Tension and compression tests were conducted on standard ASTM specimens to measure the strength of the sand-beeswax composite. The tension test specimens were fabricated as per ASTM D368-I standard. Whereas for the compression test, cylindrical specimens were cast as per ASTM D695 standard, with a specimen height of 2.54 cm and a diameter of 20 mm. In both cases, specimens were fabricated in four different mix ratios (beeswax:sand, by mass), i.e., 0.20, 0.30, 0.40, and 0.50, in three replicas each (**Figure 4**a). Tests were performed with a tensile testing machine (Zwick/Roell 5 kN Proline; Xforce P 1 kN load cell).

Samples with mixing ratios lower than 0.20 displayed an amount of molten beeswax insufficient to consistently coat the sand particles. On the contrary, at higher mixing ratios, it was challenging to maintain mixture homogeneity as the sand utilized in this study is around three times denser than beeswax (Figure 4a). The 0.30 ratio provides a better mix and consistency among the four tested ratios. With the increased beeswax proportion, the composites show an increment in tensile strength (Figure 4f). The tensile strengths of composites with mixing ratios greater than 0.22 are higher than those of pure beeswax. Tensile and compressive strengths of pure beeswax specimens align with those available in the literature.[43,44] The compressive strength of composite specimens is higher than that of pure beeswax for all the mix ratios, and this is favorable for the cylindrical extruded structure that will primarily experience compressive loads in the subsoil region.

2.3.2. Structural Strength of Extruded Channel

The strength of extruded channels has been evaluated by conducting crushing tests with both lateral and longitudinal loadings (see Video S2, Supporting Information). Specimens were prepared by following the ASTM D695 standard that determines crushing loads for hollow cylindrical structures. Specimens with equal diameter and length of 30 mm were sliced out from extruded channels using a hot blade to minimize deformations during cutting (**Figure 5**b, bottom-right). From each channel, six

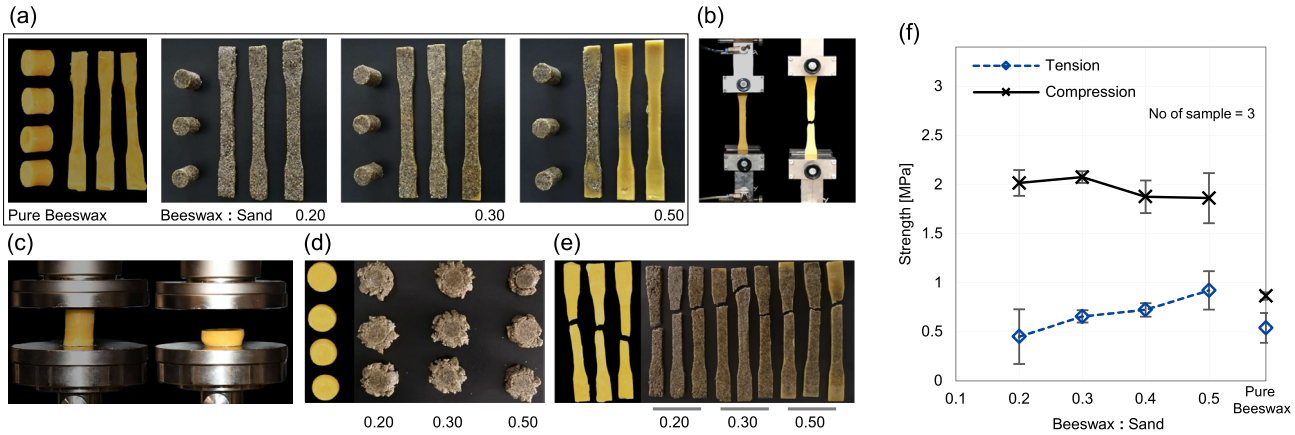

Figure 4. a) Specimens for strength testing. b) Tension test on ASTM D638-I specimens. c) Compression test on ASTM D695 specimens. d) Compression specimens after test. e) Tension specimens after test. f) Strength plots of sand-beeswax composite with different mixing ratios. The strength of pure beeswax specimens is plotted on the secondary vertical axis.

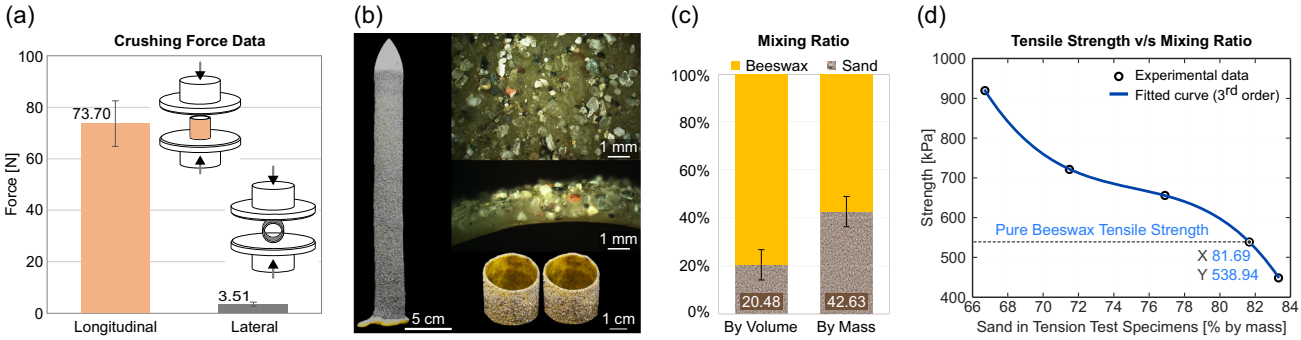

Figure 5. a) Force data from longitudinal and lateral crushing of test specimens sliced from the extruded channels. b) Snapshot of extruded channel (left), microscopy image of surface and cross-section (top-right and middle), and sample specimens for crushing test (bottom-right). c) Composition of extruded structures; sand by % volume and % mass. d) Tension test data of sand-beeswax specimens (conducted as per ASTM D368-I). The fitted curve is a third-order polynomial. The dashed line represents the strength value of pure beeswax specimens. The data label represents the composite mixing ratio corresponds to the strength of pure beeswax.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

specimens were sliced, three for longitudinal and three for lateral loading. Three channels were used for a total of nine specimens tested for each loading.

Results of the built cylindrical structures show that they can bear up to 73.70 ± 8.85 N in the longitudinal and 3.51 ± 0.82 N in the radial directions (Figure 5). These results emphasize the importance of apical growth of BeeRootBot since it permits the use of softer materials such as beeswax in structure building. If we consider a scenario where a tubular structure made of a soft material, such as beeswax, is pushed into the same soil, it may rupture due to insufficient structural strength to withstand the penetration force (penetration forces reach $\geq$100 N). However, by growing within the soil media, we can employ such materials in structural applications. Here, the strength evaluation is carried out under uniaxial loading conditions to get a rough estimate of structural strength. However, in the soil, the structures will experience triaxial loading conditions. Therefore, our results represent lower bounds of crushing loads.

2.3.3. Composition Analysis of Extruded Channel

The mixing of soil and beeswax during extrusion is crucial for the strength of the growing structure. To quantify the mixing of soil and beeswax, specimens of 90 mm height were cut from extruded channels, placed in glass-made Petri dishes, and melted inside an oven (see Figure S4, Supporting Information). The mixing ratio was estimated by equating the mass, density, and volume relations as follows:

$$V_{sd} + V_b = V_c \tag{5}$$

$$\rho_{sd}\, V_{sd} + \rho_b\, V_b = m_c \tag{6}$$

where V_{sd} is the soil volume, V_b is the beeswax volume, V_c is the total volume, ρ_{sd} is the soil density, ρ_b is the beeswax density, and m_c is the total mass. The density of pure beeswax was estimated in the same way by melting it in a Petri dish sized 27.3 cm^3, resulting in 954 kg m^3. Analogously, 45.77 g of soil was used to fill the same Petri dish. However, in this case, it was necessary to account for the air gaps between the soil grains. To address this, 11.2 mL of water was poured into the dish to fill the voids, and then the effective volume was determined by subtracting the volume of water from the total volume of the Petri dish, yielding a soil density of 2795 kg m^{-3}. Given the obtained densities, the volume of beeswax and soil from the melted samples was estimated and the corresponding mixing ratios in mass and volume were calculated. In addition, sample microscope (KH-8700, Hirox) images were captured to visually confirm the mixing results (See Figure S5, Supporting Information).

The extruded structures (an example is shown in Figure 5b) resulted in a 20.48 ± 4.95% sand by volume or 42.63 ± 7.78% sand by mass composition (Figure 5c). Microscopy images confirm the well-established bonding and the high level of mixing obtained by the beeswax with sand grains (Figure 5b, right-top & middle). The amount of beeswax consumed per unit extrusion length resulted in 80.56 ± 5.66 g m^{-1}. In other words, with one kilogram of beeswax, the probe can extrude up to a 12.40 ± 0.90 m-long structure. Due to the hydrophobic nature of beeswax, the mixing of soil and beeswax also depends on the moisture content present in the soil. For the moment, the growing tests were performed on sandy soil dried at ambient conditions. However, the hydrophobic nature provides substantial benefits on the initial design requirement of a water-resistant structure that can sustain the moisture variation in outdoor soil. The growing system can work with loose and medium-compacted silt and sandy soils, up to a particle size of $\leq$4 mm; that is twice the size of the extrusion die.

2.3.4. Binder Material Saving

From the strength analysis of standard ASTM D638-I specimens, as shown in Figure 5d, it turns out that to achieve the strength value of pure beeswax, the sand-beeswax composite requires a significantly lower amount of beeswax. The same strength can be achieved by sand-beeswax composite made with 81.69% sand by mass or 60% sand by volume. This indicates that it is possible to save up to 60% of beeswax by utilizing soil as a feedstock material while maintaining the same structural strength as pure beeswax. This highlights the usefulness of the proposed bore construction approach in space missions, where the isolation of the probe plays a crucial role, as emphasized by the Mars InSight mission.[20] The compression test data for the sand-beeswax mixture has higher strength than pure beeswax, making it unfeasible to plot the pure beeswax specimen strength on the fitted curve. Therefore, the material savings are quantified using only the tension test data. These results also confirm the limits we considered for assessing the design parameters (Figure 3a), as using ratios higher than 80% sand by mass or lower than 20% beeswax by mass depreciates the strength of composite while releasing more than 80% beeswax would lead to excessive binder consumption and possibly a thicker channel wall that could not solidify fast enough to cope with the probe advancement rate. For the extrusion die dimension (Figure 3) of 2 mm, the observed channel thickness was 1.7 ± 0.12 mm. The statistical variation among the values is $\approx$15% and is considered reasonable at such a miniaturized scale. This occurs due to the radial rebound of laterally compressed soil and the gravitational drag of molten beeswax.

2.4. Penetration Efficiency

To evaluate the penetration efficiency of our growing probe, we compared its performance with that of a nongrowing probe. A shaft of 3 cm in diameter was 3D-printed in PLA material and connected to a dummy probe, and pushed into the soil without beeswax extrusion (See Figure S6, Supporting Information). The force versus. displacement was recorded for penetrations performed with and without growing, four trials each. After each trial, the soil was rearranged by first removing about 40% of the soil from an opening located centrally at the bottom of the chamber and then refilling it by pluviation until the initial height and re-establishing approximately the same density for each trial. The penetration efficiency (η) for the growing probe was then defined as

$$\eta = \frac{F_{No-Grow} - F_{Grow}}{F_{No-Grow}} \tag{7}$$

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

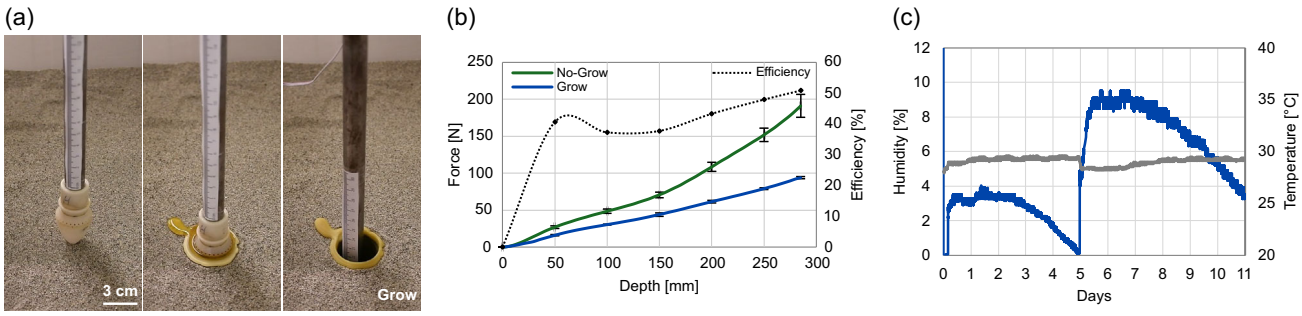

Figure 6. a) Snapshots of penetration with the growing probe. b) Penetration force data for both growing and nongrowing scenarios. The dashed curve represents the penetration force efficiency of growth over no-growth. c) Subsoil humidity and temperature data recorded by the probe over an 11-day period. Two watering instants were triggered during the acquisition period: on day one and day five.

where, F_{Grow} is the force experienced by the growing probe and $F_{No-Grow}$ is the force experienced by the nongrowing probe.

Figure 6a shows three instances (starting condition, anchoring, growing) of the penetration performed by the growing probe. The F_{Grow} accounts only for the cone resistance since the probe is the only part in contact with and moving with respect to the soil (see Figure S7, Supporting Information)

$$F_{Grow} = F_{cone_resistance} \tag{8}$$

However, the $F_{No-Grow}$ accounts for the cone resistance and bodily friction

$$F_{No-Grow} = F_{cone_resistance} + F_{friction} \tag{9}$$

with $F_{friction}$ defined as the product of lateral stress and contact surface area, both increasing with depth. Thus, eq. (7) can be expressed in subsoil resistances as

$$\eta = \frac{F_{friction}}{F_{cone_resistance} + F_{friction}} \tag{10}$$

Our results demonstrate that the growing probe is ≈40–50% more efficient than the one without growth (Figure 6b). After the first couple of centimeters, the efficiency curve has a net positive slope, meaning the bodily frictional forces are dominating over the cone resistance at greater depth. The positive slope mathematically means that with depth, the increment in $F_{friction}$ is greater than the increment in $F_{cone_resistance}$. Hence, penetration with apical growth becomes more efficient with increasing depth due to the limited movable surface area. This suggests that the proposed approach leads to significant savings in penetration force with depth, which in turn reduces operational costs: less force translates to the need for lower power machinery. Based on our tests, BeeRootBot consumes 47 kJ for the initial warmup of the syringe pump and supply channel and 100 J mm^{-1} to grow. Considering the tested growth of ≈30 cm depth with 100 N force required, we estimated a cost of transport of $15.67 \cdot 10^{-4}$ for 30 cm of displacement and a consequent energy efficiency of $6.4 \cdot 10^{-4}$. The initial energy cost becomes less significant as the distance increases, potentially improving energy efficiency. However, it must also be considered that the resistance force increases with depth. Therefore, the effective efficiency should be evaluated at a more appropriate depth, where the resistive pressure stabilizes.[25]

2.5. Remote Monitoring

As a proof-of-concept for monitoring application scenarios, we embedded in the probe a custom electronic board with digital temperature (SHT21, Sensirion AG) and color (VEML3328, Vishay Intertechnology Inc.) sensors along with a microcontroller with integrated Bluetooth low energy communication (CYBLE-014008-00, Infineon Technologies) to acquire and send data to an externally located gateway. The color sensor was used here to estimate soil humidity: the higher the light reflectance, the drier the soil.[30] The sensor was calibrated by placing 800 g of sand in a pot and acquiring readings from the tip, which was buried in the sand for 20 min at a sampling frequency of one measurement every 15 s. Acquisitions were performed in fully dry sand and at 2%, 5%, and 10% of humidity by mass. The sensing approach is analogous to the one demonstrated for sandy soils,[30] which reported the estimates achievable above 10% humidity to be inaccurate. The obtained calibration curve was a 2nd order polynomial line $y = 10^3x^2 - 0.82x + 151$. To test the monitoring application, we performed a penetration of 5 cm depth into our soil chamber and acquired the readings for about 11 consecutive days. Water was poured at two instants of time, from the top at the exterior of the generated channel: 40 mL after a few hours from the beginning of acquisition and 80 mL on the fifth day. The room was kept at a constant temperature of 30 °C and room humidity of 30% to facilitate drying. The readings were postprocessed to estimate the evolution of humidity over time.

The proposed probe can be envisioned as a borehole construction and consolidation method designed to facilitate, for instance, the insertion and retrieval of sensors, as well as the long-term monitoring of specific parameters of interest in the soil. We demonstrate the possibility of equipping the probe with sensors for remote monitoring of temperature and humidity. Figure 6c presents an 11 day data acquisition. The two instances of water pouring are clearly identifiable and we can observe how humidity and temperature vary with opposite trends but remain synchronized.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3. Conclusion

This article introduces a soil-intruding probe that imitates the apical growth of plant roots by circumstantially injecting a bio-based binder from its apical extremity and harnesses the local in situ soil to construct and consolidate a borehole while advancing in depth. This penetration strategy limits the resistive stresses to only the cone resistance, as there is no other movable body part in contact with the soil, and consequently, there is no relative friction between the soil and the intruding shaft. This reduces the penetration force requirements and, in turn, the required power while increasing efficiency with depth compared to a similar intruder that does not construct a soil-reinforcing channel on the exterior of its tip.

This technique is minimally invasive, as no soil excavation is involved. On the contrary, the soil becomes part of the constructed body, which additionally enhances structural strength and reduces binder consumption, as the strength of the sand-beeswax composite is higher than that of the matrix material, i.e., the binder (beeswax). Combining soil with a bio-based, biodegradable binder also promotes a more sustainable approach to small-scale borehole construction and lining, making it more compatible with surrounding ecosystems. The biodegradable structure, along with a probe also made of biodegradable polymer (we tested PHA as a replacement for SLA resin), and retraction of most of the robotic components, can provide a sustainable solution that limits the dispersion of contaminating elements into the environment. This approach is particularly convenient for subsoil exploration and monitoring applications.

The constructed channel provides a stable and environmentally friendly passage for deploying sensors, transporting matrix material and other components at depth, and maintaining communication with the surface host. In the context of state-of-the-art tunneling technologies, which are based on discrete stepwise boring and reinforcement operations, we are proposing a continuous tunneling approach that reduces process time, material usage, and costs. In the BeeRootBot setup, tip advancement and binding material release systems are decoupled, unlike other growing mechanisms.[27,32] This allows for faster penetration speed with respect to previous solutions. In geotechnical applications, it can be employed as a convenient strategy for directly measuring cone penetration resistance.

Harnessing in situ local resources and binder material-saving features underscore the suitability of this approach in space applications. We employ sandy soil in this study, which resembles simulants for Martian and Lunar soil. The probe offers scope for sensor integration and could be utilized as a carrier for sensor deployment in subsoil monitoring. In ecological applications, the borehole could be used to provide isolation, protecting seedlings during the rehabilitation of toxic environments.[45,46] The borehole could serve as an intervention channel for improving the health of stressed ecosystems by supplying nutrients and biological growth elements. Overall, its features benefit both on-Earth and outer space applications, reducing mission costs and space requirements.

4. Experimental Section

Soil Chamber and Penetration Test Protocol: A cubical container was fabricated to host the soil and test the functionalities and penetration capability of the overall system. To limit border effects, the container was dimensioned $\geq$16 times bigger than the probe diameter, reaching a width of 50 cm. We employed a granular media (sand with particle size $\leq$1 mm from Leroy Merlin, local distributor). This chamber was employed during iterative design experiments, penetration tests, and to produce sample structures for their characterization.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation program (grant 101003304) (I-WOOD). This work is also part of the "Technologies for Sustainability'' Flagship program of the Italian Institute of Technology (IIT).

Open access publishing facilitated by Istituto Italiano di Tecnologia, as part of the Wiley - CRUI-CARE agreement.

Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Sachin Sachin: conceptualization (lead); data curation (lead); formal analysis (lead); investigation (lead); methodology (lead); software (lead); validation (lead); visualization (lead); writing—original draft (lead). **Alessio Mondini**: conceptualization (supporting); methodology (supporting); resources (supporting); software (lead); validation (supporting); writing—original draft (supporting); writing—review & editing (supporting). **Stefano Mariani**: methodology (supporting); resources (supporting). **Emanuela Del Dottore**: Conceptualization (Supporting); Investigation (Supporting); Methodology (Supporting); project administration (supporting); supervision (supporting); validation (supporting); visualization (supporting); writing—original draft (supporting); writing—review & editing (supporting). **Barbara Mazzolai**: conceptualization (lead); funding acquisition (lead); methodology (lead); project administration (lead); resources (lead); supervision (lead); writing—original draft (supporting); writing—review & editing (supporting).

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

bio-based binder, biodegradable structure, bioinspiration, growing robots, plant root-inspired, remote monitoring, subsoil exploration

Received: July 2, 2025
Published online: September 2, 2025

[1] Y. Bar-Cohen, K. Zacny, *Drilling in Extreme Environments: Penetration and Sampling on Earth and Other Planets*, John Wiley & Sons **2009**.
[2] N. R. Council, D. on Earth, L. Studies, C. on Geosciences, Resources, C. on Advanced Drilling Technologies, *Drilling and Excavation Technologies for the Future*, National Academies Press **1994**

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[3] A. Martinez, J. DeJong, I. Akin, A. Aleali, C. Arson, J. Atkinson, P. Bandini, T. Baser, R. Borela, R. Boulanger, et al., *Géotechnique* **2022**, *72*, 687.
[4] W. Zhang, J. Xiang, R. Huang, H. Liu, *Biogeotechnics* **2023**, *1*, 100025.
[5] K. Nagaoka, T. Kubota, M. Otsuki, S. Tanaka, in *2008 IEEE/RSJ International Conference on Intelligent Robots and Systems*, IEEE **2008**, pp. 4104–4109.
[6] A. Koller-Hodac, D. P. Germann, A. Gilgen, K. Dietrich, M. Hadorn, W. Schatz, P. E. Hotz, in *2010 IEEE International Conference on Robotics and Automation*, IEEE **2010**, pp. 1209–1214.
[7] R. A. Russell, *Adv. Robot.* **2011**, *25*, 119.
[8] K. Zacny, D. Currie, G. Paulsen, T. Szwarc, P. Chu, *Planet. Space Sci.* **2012**, *71*, 131.
[9] H. Wei, Y. Zhang, T. Zhang, Y. Guan, K. Xu, X. Ding, Y. Pang, *Space Sci. Rev.* **2021**, *217*, 1.
[10] K. M. Dorgan, K. A. Daltorio, *Front. Robot. AI* **2023**, *10*, 1057876.
[11] H. Bagheri, D. Stockwell, B. Bethke, N. K. Okwae, D. Aukes, J. Tao, H. Marvi, *Acta Geotech.* **2024**, *19*, 1435.
[12] A. Menciassi, D. Accoto, S. Gorini, P. Dario, *Auton. Robot.* **2006**, *21*, 155.
[13] B. H. Shin, S.-W. Choi, Y.-B. Bang, S.-Y. Lee, *Smart Mater. Struct.* **2011**, *20*, 105020.
[14] T. Nakatake, M. Konno, A. Mizushina, Y. Yamada, T. Nakamura, T. Kubota, in *2016 IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*, IEEE **2016**, pp. 407–412.
[15] K. Isaka, K. Tsumura, T. Watanabe, W. Toyama, M. Sugesawa, Y. Yamada, H. Yoshida, T. Nakamura, *IEEE Access* **2019**, *7*, 103127.
[16] Y. Ozkan-Aydin, B. Liu, A. C. Ferrero, M. Seidel, F. L. Hammond, D. I. Goldman, *Bioinspir. Biomimet.* **2021**, *17*, 016001.
[17] R. Niiyama, K. Matsushita, M. Ikeda, K. Or, Y. Kuniyoshi, *Soft Matter* **2022**, *18*, 417990.
[18] R. Das, S. P. M. Babu, F. Visentin, S. Palagi, B. Mazzolai, *Sci. Rep.* **2023**, *13*, 11571.
[19] M. Barenboim, A. Degani, in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE **2020**, pp. 829–835.
[20] T. Spohn, T. L. Hudson, L. Witte, T. Wippermann, L. Wisniewski, B. Kedziora, C. Vrettos, R. D. Lorenz, M. Golombek, R. Lichtenheldt, et al., *Adv. Space Res.* **2022**, *69*, 3140.
[21] N. D. Naclerio, C. M. Hubicki, Y. O. Aydin, D. I. Goldman, E. W. Hawkes, in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE **2018**, pp. 5918–5923.
[22] N. D. Naclerio, A. Karsai, M. Murray-Cooper, Y. Ozkan-Aydin, E. Aydin, D. I. Goldman, E. W. Hawkes, *Science Robotics* **2021**, *6*, eabe2922.
[23] E. Del Dottore, A. Sadeghi, A. Mondini, V. Mattoli, B. Mazzolai, *Front. Robot. AI* **2018**, *5*, 16.
[24] A. Sadeghi, A. Tonazzini, L. Popova, B. Mazzolai, *PloS One* **2014**, *9*, e90139.
[25] S. R. Pirrone, E. Del Dottore, L. Sibille, B. Mazzolai, *IEEE Robot. Autom. Lett.* **2023**, *8*, 3438.
[26] B. Mazzolai, A. Mondini, P. Corradi, C. Laschi, V. Mattoli, E. Sinibaldi, P. Dario, *IEEE/ASME Trans. Mechat.* **2010**, *16*, 201.
[27] A. Sadeghi, A. Mondini, B. Mazzolai, *Soft Robot.* **2017**, *4*, 211.
[28] D. Lunni, E. Del Dottore, A. Sadeghi, M. Cianchetti, E. Sinibaldi, B. Mazzolai, in *Biomimetic and Biohybrid Systems: 7th International Conference, Living Machines 2018, Paris, France, July 17–20, 2018, Proceedings 7*, Springer **2018**, pp. 288–299.
[29] A. Sadeghi, E. Del Dottore, A. Mondini, B. Mazzolai, *Soft Robot.* **2020**, *7*, 85.
[30] E. Del Dottore, A. Mondini, D. Bray, B. Mazzolai, in *Conference on Biomimetic and Biohybrid Systems*, Springer **2023**, pp. 184–196.
[31] B. Solanke, F. Onita, O. Ochulor, H. Iriogbe, *Eng. Sci. Technol. J.* **2024**, *5*, 2737.
[32] E. Del Dottore, A. Mondini, N. Rowe, B. Mazzolai, *Sci. Robot.* **2024**, *9*, eadi5908.
[33] N. Hawaldar, J. Zhang, *Int. J. Adv. Manuf. Technol.* **2018**, *97*, 1037.
[34] M. V. Mohiuddin, S. F. Hussainy, A. Krishnaiah, P. Laxminarayana, *Int. J. Adv. Manuf. Technol.* **2017**, *90*, 3147.
[35] T. Sivarupan, N. Balasubramani, P. Saxena, D. Nagarajan, M. El Mansori, K. Salonitis, M. Jolly, M. S. Dargusch, *Addit. Manuf.* **2021**, *40*, 101889.
[36] D. B. Gilmer, L. Han, M. L. Lehmann, D. H. Siddel, G. Yang, A. U. Chowdhury, B. Doughty, A. M. Elliott, T. Saito, *Nat. Commun.* **2021**, *12*, 1.
[37] M. Upadhyay, T. Sivarupan, M. El Mansori, *J. Manuf. Process.* **2017**, *29*, 211.
[38] R. Ramakrishnan, B. Griebel, W. Volk, D. Günther, J. Günther, *Adv. Mater. Res.* **2014**, *1018*, 441.
[39] A. O. Hanstveit, *Chemosphere* **1992**, *25*, 605.
[40] R. Buchwald, M. D. Breed, A. R. Greenberg, *J. Exper. Biol.* **2008**, *211*, 121.
[41] J.-P. Verbelen, T. D. Cnodder, J. Le, K. Vissenberg, F. Baluška, *Plant Signal. Behav.* **2006**, *1*, 296.
[42] A. K. Mishra, F. Tramacere, R. Guarino, N. M. Pugno, B. Mazzolai, *PLoS One* **2018**, *13*, e0197411.
[43] J. Morgan, S. Townley, G. Kemble, R. Smith, *Mater. Sci. Technol.* **2002**, *18*, 463.
[44] M. E. Hossain, C. Ketata, M. R. Islam, in *Proceeding of Third International Conference on Modeling, Simulations and Applied Optimization, Sharjah, United Arab Emirates* **2009**, pp. 1–5.
[45] Y. L. de Paula, R. R. de Melo, E. A. d. O. Paula, E. R. C. de Morais, F. Rodolfo Junior, A. S. Pimenta, R. R. A. de Oliveira, J. A. G. de Souza, M. V. Scatolino, T. D. Pedrosa, *Waste Biomass Valoriz.* **2024**, *15*, 697.
[46] R. Moradi, N. Pourghasemian, M. Naghizadeh, *J. Clean. Prod.* **2019**, *229*, 1251.

© 2025 The Author(s). Advanced Intelligent Systems published by Wiley-VCH GmbH

26404567, 2026, 2, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500720, Wiley Online Library on [20/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License